GenMSLnet: Generative Modality-Specific Learning-Enhanced Collaborative Fusion Network for Hyperspectral and SAR Joint Classification

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***Abstract*—Hyperspectral (HS) images and synthetic aperture radar (SAR) data provide complementary information for land cover interpretation. However, their cross-modal collaboration remains insufficient in multisatellite observation systems. Moreover, both remote sensing (RS) modalities are sensitive to environmental interference, hindering critical feature acquisition. To overcome these challenges, a generative modality-specific learning-enhanced collaborative fusion network called GenMSLnet is proposed for HS and SAR joint classification. Our core innovation is modality-specific feature learning, directly guided by the RS inherent properties. The framework couples a multiscale dual-stream generative adversarial network (MDGAN) for modality-specific feature learning with a transformer-based dual-path cross-attention (TDCA) network for complementary fusion. MDGAN comprises dual streams for HS image (HSI) and SAR data. The former captures spectral–spatial details by multiscale feature extraction and attention, while the latter employs orientation-aware sampling to capture geometry- and orientation-sensitive backscattering signals. A residual separable U-Net discriminator (RSU-NetD) incorporating depthwise separable channel attention residual blocks (DSCA ResBlocks) is considered to strengthen feature discrimination. TDCA balances RS modality contributions by introducing a novel gated bidirectional dynamic cross-attention mechanism. Experiments demonstrate that GenMSLnet outperforms state-of-the-art techniques quantitatively and qualitatively. Codes will be available at https://github.com/zhanghongzhan-123/GenMSLnet**

***Index Terms*—Generative model, hyperspectral (HS), joint classification, modality specific, synthetic aperture radar (SAR).**

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I. Introduction

THE rapid development of remote sensing (RS) technology has revolutionized natural resource management by providing unprecedented data analysis capabilities [1]. hyperspectral images (HSIs) and synthetic aperture radar (SAR) data offer complementary properties that are critical for agricultural management, environmental assessment, urban planning, etc. [2]. HSIs capture spectral–spatial characteristics by hundreds of contiguous narrow bands [3]. However, these images are extremely sensitive to atmospheric conditions, e.g., cloud cover and spatial resolution, which can obscure small-scale boundaries. On the contrary, SAR utilizes active microwave imaging technologies, which enable continuous day-and-night observation regardless of weather conditions. This observation paradigm makes SAR data indispensable for forest monitoring [4]. However, SAR data suffer from inherent speckle noise and limited spectral discriminability [5]. To overcome these limitations, comprehensively fusing HSIs and SAR data has become increasingly urgent [6]. Existing approaches consist of pixel-level, feature-level, and decision-level fusion.

Pixel-level fusion operates on raw data by combining or replacing pixels across modalities to generate high-quality synthesized images. Kulkarni and Rege [7] conducted a systematic review of pixel-level fusion strategies. Li et al. [8] proposed a probabilistic fusion approach by combining pixel- and superpixel-level classifiers to analyze multisource information. Liu et al. [9] proposed a convolutional neural network (CNN) enhanced graph convolution network to bridge Euclidean and non-Euclidean representations. Nartey et al. [10] combined pixel-level covariance pooling features with superpixel-level graph representations. These technologies demonstrate that pixel-level fusion effectively enhances spatial resolution and enriches image details. Nevertheless, compared with feature-level fusion, it tends to involve higher computational complexity and greater sensitivity to noise.

Feature-level fusion combines high-level features from RS modalities, focusing on preserving information while suppressing noise interference. Zhong et al. [11] proposed a hybrid

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bilateral-attention generative adversarial network (GAN) to fuse hierarchical global and local features for seismic image super-resolution. Hong et al. [12] proposed invariant attribute profiles to extract features robust to local semantic changes. In addition, the authors further introduced a shared-specific feature learning model [13]. Kong et al. [14] developed a dense U-shaped GAN to preserve optical and SAR texture features. Wei et al. [15] proposed an illumination-insensitive modalitywise assimilation guided multistage interaction network to address spectral distortion. Hou et al. [16] introduced a dual-channel pulse-coupled neural network that achieves enhanced feature extraction in the nonsubsampled shearlet transform domain. These methods demonstrate that feature-level fusion balances information preservation and fusion flexibility, offering improved performance while maintaining efficiency.

Decision-level fusion focuses on voting across different classifiers to obtain optimized results. Generally, multiple classifiers with a unique voting strategy are considered for different RS modalities. Zhong et al. [17] proposed an optimal decision-level fusion method based on adaptive differential evolution, which optimizes classifier weights and combines RS modalities, effectively improving classification performance. Jia et al. [18] proposed a multifeature superpixel-level decision fusion method by considering superpixel-guided kernel principal component analysis and weighted voting strategies. Generally, decision-level fusion should be aligned with feature-level fusion for satisfactory performance [19].

Fusing HSIs and SAR data still faces two critical challenges.

1) *Modality-Specific Feature Acquisition:* Conventional approaches generally rely on fixed receptive fields and localized convolution operations, struggling to acquire modality-specific features from HSIs with abundant spectral bands and SAR data with complex spatial textures. Consequently, their performance is limited in complex scenarios and real-world tasks.

2) *Heterogeneous Cross-Modal Feature Integration:* Existing approaches often depend on naive concatenation or linear weighting without adequately accounting for sensing-mechanism discrepancy and distributional differences. Moreover, many approaches lack systematic mechanisms to model cross-modal interactions, resulting in suboptimal integration of fine-grained details with global information.

To tackle these challenges, a generative modality-specific learning-enhanced collaborative fusion network called GenMSLnet is proposed. Its core innovation is developing modality-specific feature learning strategies for both modalities and incorporating their distinct capturing properties for information interaction. GenMSLnet fully exploits the complementary strengths of HSIs and SAR data through an advanced framework at multiple levels. Notably, a multiscale dual-stream generative adversarial network (MDGAN) serves as the backbone to enhance modality-specific feature representation and cross-modal distribution alignment. Both the generator and discriminator in MDGAN adopt dual-stream architectures for both specific modalities. The first stream exploits comprehensive spectral–spatial features from HSIs, while the second stream captures SAR-specific signatures, e.g., geometric, orientation-aware, and structural clues. A residual separable U-Net discriminator (RSU-NetD) is adopted as the discriminator, which integrates depthwise separable channel attention residual blocks (DSCA ResBlocks) and multilevel gradient propagation to preserve HSI spectral fidelity and suppress SAR noise. These structures further enhance cross-modal alignment. To bridge the heterogeneous distributions between modalities, GenMSLnet introduces a gated bidirectional dynamic cross-attention mechanism. This mechanism captures spectral-texture dependencies adaptively and facilitates selective integration of complementary information from both RS modalities. The contributions are provided as follows.

1) A unified end-to-end framework called GenMSLnet is proposed for hyperspectral (HS) and SAR joint classification through cross-modal representation alignment. This framework systematically extracts modality-specific features guided by RS inherent properties for heterogeneous cross-modal feature integration to enhance multisource collaboration. GenMSLnet integrates MDGAN for modality-specific adversarial feature learning and transformer-based dual-path cross-attention (TDCA) for complementary fusion. MDGAN utilizes two separate modality-specific streams, paired with an RSU-NetD and regularized to mitigate kernel redundancy and stabilize training.
2) GenMSLnet presents two modality-aware dual-stream generators aligned with the physical and statistical characteristics of each modality. The HS stream combines multiscale orientation-enhanced convolutions, a two-level residual pyramid, and lightweight channel attention to preserve spectral–spatial details. The SAR stream incorporates deformable sampling, explicit orientation reinforcement, multiscale energy-context enhancement, and spatial attention to capture geometry- and orientation-sensitive backscattering signals.
3) GenMSLnet presents TDCA to balance RS modality contributions by introducing a novel gated bidirectional dynamic cross-attention mechanism. Heterogeneous features are comprehensively aligned to yield robust and discriminative joint representations. Experiments on publicly available datasets prove that GenMSLnet outperforms comparison techniques qualitatively and quantitatively.

Section II provides some related work. Section III describes GenMSLnet in detail. Section IV provides and analyzes the experimental results. Section V concludes this work.

II. Related Works

A. HS and SAR Joint Classification

Fusing HSIs and SAR data combines rich HS spectral–spatial information with superior SAR spatial-structural representation. Early approaches primarily rely on linear feature-level strategies, e.g., principal component analysis [20]. However, due to complex nonlinear semantic cross-modal differences, these approaches tend to suffer from limited adaptability in practical applications. Xu et al. [21] proposed a contrastive learning-enhanced network for hierarchical feature interaction and adaptive fusion. Yu et al. [22] addressed modality heterogeneity in unsupervised

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recognition by an intuitive assimilation modality, guiding cross-modal subspace clustering to improve performance. Lin et al. [23] incorporated a dynamic routing mechanism for adaptive multimodal feature fusion, boosting classification performance.

Attention mechanisms have gained prominence in cross-modal modality fusion. Yang et al. [24] introduced a cross-modal Transformer attention structure to mitigate high dimensionality challenges. Li et al. [25] integrated global attention and distance suppression mechanisms to enhance multiscale feature discriminability in class-imbalanced scenarios. Li et al. [26] employed feature reconstruction embedding to align semantics and achieve complementary fusion, further strengthening the representational capability.

Despite these advances, most existing approaches rely on static attention mechanisms and lack the flexibility to modulate cross-modal feature contributions based on inputs. To address this, GenMSLnet introduces a gated bidirectional dynamic cross-attention mechanism in a GAN-based backbone architecture. This structure promotes sample-specific feature importance adjustment and enhances classification performance, further motivating the construction of TDCA in Section III.

B. GAN and Its Application in RS Fields

GANs have achieved remarkable success in analyzing high-dimensional data, with wide applications in domain translation and data synthesis [27]. Generally, a typical GAN consists of one generator G and one discriminator D with the objective $\min_G \max_D \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}(\mathbf{x})}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))]$, where p_{data} and p_z denote the distributions of real data $\mathbf{x}$ and noise $\mathbf{z}$, respectively. To address modality gaps, cross-modal GANs, e.g., CycleGAN [28] and DualGAN [29], learn bidirectional mappings between heterogeneous domains in an unsupervised manner. These techniques consider a cycle-consistency loss $\mathcal{L}_{\text{cyc}}$ as follows:

$$\begin{aligned}\mathcal{L}_{\text{cyc}} = &\mathbb{E}_{\mathbf{x}\sim p_{\text{data}}(\mathbf{x})}\|F(G(\mathbf{x})) - \mathbf{x}\|_1 \\ &+ \mathbb{E}_{\mathbf{y}\sim p_{\text{data}}(\mathbf{y})}\|G(F(\mathbf{y})) - \mathbf{y}\|_1 \end{aligned} \tag{1}$$

where G and F are different generators, $\|\cdot\|_1$ denotes the ℓ_1 norm. GAN and its variants have been widely used to generate modality-translated representations. Hong et al. [30] proposed a graph-aware cross-modal GAN to learn spectral–spatial-structural features. However, conventional GAN-based methods always suffer from training instability and semantic misalignment when facing modality heterogeneity and spectral–spatial distortion. These techniques focus on pixel/distribution-level alignment without enforcing semantic consistency, limiting their effectiveness in classification.

To address these problems, modality-specific branches and auxiliary classifiers are considered. For instance, auxiliary GAN [31] extends the discriminator by jointly predicting source probability and class labels. Yu et al. [32] developed a dual-stream GAN variant to extract both static and dynamic spectral–spatial features. Meanwhile, a dual-generator architecture incorporates modality-specific encoders and feature-level discriminators to preserve semantic information [33]. Motivated by them, a novel multibranch GAN backbone architecture is presented to integrate dual discriminators and dual generators for capturing HS and SAR features comprehensively. GenMSLnet preserves modality-specific structures and collaboratively aligns modality-shared features.

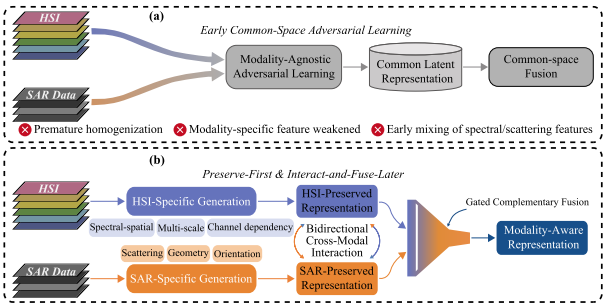


Fig. 1. Comparison between representative modality-agnostic GAN-based fusion and the modality-specific generative learning method GenMSLnet. (a) Representative modality-agnostic GAN-based fusion. (b) GenMSLnet: modality-specific generative learning.

C. Compared With GAN-Based Multimodal Fusion Networks

Many existing GAN-based multimodal fusion networks leverage adversarial learning for data augmentation, cross-modal translation, or common representation learning (RepL). Although such strategies enhance multimodal feature learning, typical modality-agnostic designs tend to project heterogeneous modalities onto a shared or unified latent representation without sufficiently preserving modality-specific features. For HSI-SAR joint classification, this practice proves inadequate, as the two modalities originate from distinct feature formation mechanisms. In particular, HSI provides spectral–spatial continuity and spectral–channel dependencies, whereas SAR data capture scattering responses, geometric structures, and orientation-sensitive textures. This motivation aligns with recent multimodal learning theory, identifying modality heterogeneity, cross-modal connections, and interactions as foundational principles in multimodal RepL [34].

Fig. 1 illustrates that the primary distinction lies in the overall modality-aware learning route rather than the use of individual modules. Typical modality-agnostic approaches tend to project HSI and SAR data into a common adversarial processor or a shared latent representation at an early stage. On the contrary, GenMSLnet adopts a *Preserve-First & Interact-and-Fuse-Later* paradigm. In particular, it first learns HSI- and SAR-specific generative representations based on their respective modality characteristics, then captures their complementary relations through bidirectional cross-modal attention, and finally aggregates useful information through gated complementary fusion.

From an inductive bias perspective, early common-space adversarial learning may lead to insufficient preservation of modality-specific discriminative features, because the HSI spectral–spatial continuity and SAR scattering–structural responses follow distinct data characteristics. Rather than directly homogenizing the two modalities, GenMSLnet first enhances them by modality-specific generative branches, followed by cross-modal interaction and adaptive fusion. This

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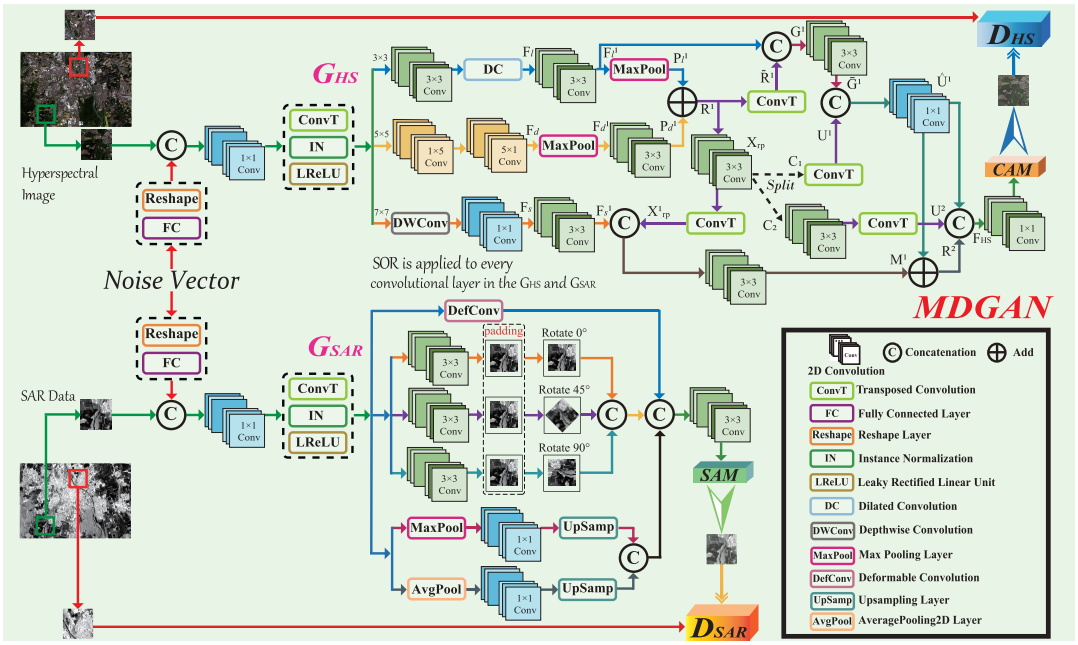


Fig. 2. Scheme of the MDGAN generators in GenMSLnet for feature extraction, where D_{HS} and D_{SAR} are the MDGAN discriminators in GenMSLnet.

TABLE I

NOVELTY COMPARISON BETWEEN REPRESENTATIVE GAN-BASED FUSION METHODS AND GENMSLNET

Aspect	Representative GAN-based Fusion Methods	GenMSLnet
Learning Paradigm	Usually following a *Homogenize-then-Fuse* paradigm through early shared or unified latent representation	Following a *Preserve-First & Interact-and-Fuse-Later* paradigm through modality-specific generative RepL before cross-modal interaction and fusion
Heterogeneity Treatment	Possibly mixing heterogeneous modalities before sufficiently preserving modality-specific features	Separately preserving HSI spectral-spatial features and SAR scattering-structural features
GAN Role	Mainly used for generation, translation, augmentation, or common-space RepL	Used to enhance modality-specific discriminative representations for HSI and SAR data
Cross-modal Relation	Often relying on shared representation, concatenation, or direct fusion	Using bidirectional cross-modal interaction after modality-specific RepL
Fusion Mechanism	Fusing common or enhanced features directly	Using gated complementary fusion to adaptively integrate useful HSI/SAR information
Core Novelty	Improving multimodal learning mainly through adversarial enhancement or common representation learning	Shifting the overall learning route from *Homogenize-then-Fuse* to *Preserve-First & Interact-and-Fuse-Later*

design makes the fusion process more compatible with the heterogeneous nature of HSI-SAR joint classification.

Therefore, the key novelty of GenMSLnet lies not in the isolated use of GANs, attention mechanisms, or gating modules, but rather in the modality-aware learning route imposed on heterogeneous HSI-SAR data. HSI and SAR data are first represented in a modality-specific manner and subsequently integrated through cross-modal interaction and gated complementary fusion. This explains why the proposed architecture is distinct from a naive combination of existing modules.

Table I summarizes the key distinctions between representative GAN-based fusion methods and GenMSLnet. In contrast to common-space adversarial fusion, GenMSLnet explicitly accounts for the distinct characteristics, rendering the learning process more consistent with the HSI-SAR heterogeneity.

Collectively, GenMSLnet chooses a modality-aware learning route from *Homogenize-then-Fuse* to *Preserve-First & Interact-and-Fuse-Later*. At the representation-learning level, this route is implemented by shifting the feature-learning

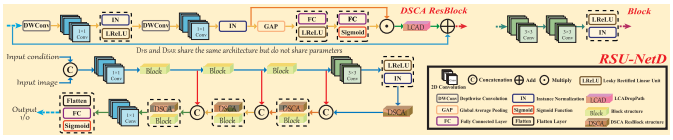


Fig. 3. Scheme of the MDGAN discriminators called RSU-NetD in GenMSLnet for feature extraction.

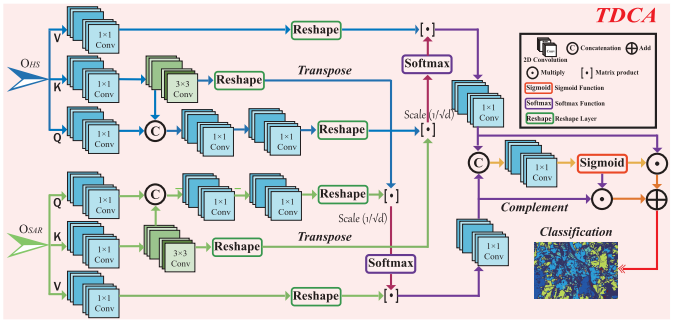


Fig. 4. Scheme of TDCA in GenMSLnet for complementary fusion, where Q, K, and V are query, key, and value representations.

strategy from modality-agnostic common RepL to modality-specific generative RepL. Such a design provides a task-oriented inductive bias that better matches the heterogeneous characteristics of HSI and SAR data, thereby clarifying the scientific motivation and architectural novelty of the proposed modality-specific learning framework.

III. METHODOLOGY

The structure schemes are shown in Figs. 2–4. GenMSLnet comprises a multiscale dual-stream GAN (MDGAN) for feature extraction and a TDCA network for complementary fusion. Given significant differences in imaging mechanisms, using a single generator within a unified parameter space to

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approximate both heterogeneous distributions leads to distortion. To address this, MDGAN employs two modality-specific generator branches, i.e., G_{HS} and G_{SAR}. G_{HS} learns continuous HS spectral responses and subtle land cover variations, while G_{SAR} focuses on SAR geometric structures. Each generator branch is paired with a dedicated discriminator, denoted as D_{HS} and D_{SAR}.

A. Generators in MDGAN

1) Separated Orthogonal Regularizer: The structure in MDGAN is shown in Fig. 2. Multiple overly correlated convolutional kernels weaken the RS information extraction. In GenMSLnet, a separated orthogonal regularizer (SOR) is introduced in the generators G_{HS} and G_{SAR} to encourage diverse and complementary kernels. For one convolutional layer weight W, it is reshaped into $\mathbb{R}^{C_{\mathrm{out}}\times(C_{\mathrm{in}}k^2)}$ by unfolding the spatial and channel dimensions, where C_{in} and C_{out} are the input and output channel numbers, k is the kernel size (unless otherwise specified, $k = 3$). The corresponding Gram matrix $\mathbf{K} = WW^{\top}$ is regularized as

$$\mathcal{L}_{\mathrm{SOR}} = \lambda_1 \|\mathrm{diag}\,(\mathbf{K}) - \mathbf{1}\|_2^2 + \lambda_2 \|\mathbf{K} \odot (\mathbf{1} - I)\|_1 \tag{2}$$

where I is the identity matrix, $\odot$ is the Hadamard product, λ_1 and λ_2 are coefficients, and $\|\cdot\|_1$ and $\|\cdot\|_2$ are the ℓ_1 and ℓ_2 norms, respectively. Equation (2) consists of two terms: the first term normalizes a unit-norm constraint on each filter by aligning the diagonal of $\mathbf{K}$ toward 1; the second term promotes filter diversity by sparsifying off-diagonal entries toward 0 using an ℓ_1 penalty. This strategy encourages kernelwise orthogonality, yielding complementary features for cross-modal fusion.

2) G_{HS}: G_{HS} combines a multiscale orientation-enhanced module with a lightweight channel attention mechanism. For one HSI input $\mathbf{X}_{\mathrm{HS}} \in \mathbb{R}^{H\times W\times C_\lambda}$, where H, W, and C_λ are the height, width, and number of spectral channels, the multiscale orientation-enhanced module extracts complementary spectral–spatial features through parallel branches with a 3×3 dilated convolutional layer, a 1×5 followed by 5×1 asymmetric convolutional layer pair, and a 7×7 depthwise separable convolutional layer. Pyramid downsampling and upsampling, along with residual concatenation, are considered to capture multiresolution contextual information. Finally, a lightweight channel attention mechanism is applied to the C_λ-channel projected output to adaptively reweight spectral responses across channels and preserve spectral details.

a) Triple-branch convolution: G_{HS} adopts a three-branch parallel structure. The first branch captures local contexts by a 3×3 convolutional layer followed by a 3×3 dilated convolutional layer (the dilation rate is 2), generating $\mathbf{F}_l \in \mathbb{R}^{H\times W\times 32}$. The second branch extracts directional textures with sequential 1×5 and 5×1 convolutions, yielding $\mathbf{F}_d \in \mathbb{R}^{H\times W\times 32}$. The third branch approximates nonlocal spectral operations using a 7×7 depthwise separable convolutional layer and a 1×1 pointwise convolutional layer, generating $\mathbf{F}_s \in \mathbb{R}^{H\times W\times 32}$.

b) Two-level residual pyramid: A two-level residual pyramid integrates multiscale information with error correction in GenMSLnet, replacing direct branch concatenation. It combines max-pooled features with stride-aligned residual connections to form a composite representation, which is then channel-split and upsampled by transposed convolutional layers. This strategy promotes unimpeded gradient propagation and enhances cross-scale contextual integration.

Initially, a 3×3 convolutional layer is applied to $\mathbf{F}_l$ for $\mathbf{F}_l^1 \in \mathbb{R}^{H\times W\times 64}$, followed by 2×2 max pooling for $\mathbf{P}_l^1 \in \mathbb{R}^{H/2\times W/2\times 64}$. In parallel, $\mathbf{F}_d$ is processed by 2×2 max pooling followed by a 3×3 convolutional layer for $\mathbf{P}_d^1 \in \mathbb{R}^{H/2\times W/2\times 64}$. $\mathbf{P}_l^1$ and $\mathbf{P}_d^1$ are combined as $\mathbf{R}^1 = \mathbf{P}_l^1 + \mathbf{P}_d^1$. Subsequently, $\mathbf{R}^1$ is upsampled to the high-resolution scale by a transposed convolutional layer (stride is 2 and padding is valid) for $\tilde{\mathbf{R}}^1$, which is then fused with $\mathbf{F}_l^1$ by concatenation for $\mathbf{G}^1 \in \mathbb{R}^{H\times W\times 128}$. A subsequent 3×3 convolutional layer is applied to $\mathbf{G}^1$ for $\overline{\mathbf{G}}^1 \in \mathbb{R}^{H\times W\times 64}$. Meanwhile, $\mathbf{R}^1$ is transformed by a 3×3 convolutional layer into $\mathbf{X}_{\mathrm{rp}} \in \mathbb{R}^{H/2\times W/2\times 64}$ and evenly split along the channel dimension into two tensors, i.e., $\mathbf{C}_1, \mathbf{C}_2 \in \mathbb{R}^{H/2\times W/2\times 32}$. $\mathbf{C}_1$ is upsampled by a transposed convolutional layer (stride is 2) for $\mathbf{U}^1 \in \mathbb{R}^{H\times W\times 64}$, concatenated with $\overline{\mathbf{G}}^1$, and compressed by a 1×1 convolutional layer for $\widehat{\mathbf{U}}^1 \in \mathbb{R}^{H\times W\times 64}$ in turn. $\mathbf{C}_2$ is refined by a 3×3 convolutional layer and upsampled in the same manner for $\mathbf{U}^2 \in \mathbb{R}^{H\times W\times 64}$.

c) High-level feature fusion: Notably, $\mathbf{F}_s$ is processed by a 3×3 convolutional layer for $\mathbf{F}_s^1 \in \mathbb{R}^{H\times W\times 64}$ along the spectral channel. In parallel, $\mathbf{X}_{\mathrm{rp}}$ is upsampled by a transposed convolutional layer for $\mathbf{X}_{\mathrm{rp}}^1 \in \mathbb{R}^{H\times W\times 64}$ and then concatenated with $\mathbf{F}_s^1$. The output is processed by a 3×3 convolutional layer for $\mathbf{M}^1 \in \mathbb{R}^{H\times W\times 64}$. Then, $\mathbf{M}^1$ and $\widehat{\mathbf{U}}^1$ are added for $\mathbf{R}^2 \in \mathbb{R}^{H\times W\times 64}$. Finally, $\widehat{\mathbf{U}}^1$, $\mathbf{U}^2$, and $\mathbf{R}^2$ are concatenated along the channel dimension as

$$\mathbf{F}_{\mathrm{HS}} = \left[\widehat{\mathbf{U}}^1 \parallel \mathbf{U}^2 \parallel \mathbf{R}^2\right] \in \mathbb{R}^{H\times W\times 192} \tag{3}$$

where $\parallel$ denotes the channelwise concatenation. The fused features are projected to C_λ channels using a 1×1 convolutional layer for $\tilde{\mathbf{F}}_{\mathrm{HS}} \in \mathbb{R}^{H\times W\times C_\lambda}$. While this structure facilitates bottom-up propagation and multiscale consistency, channelwise spectral imbalance may persist in the output despite the redundancy mitigation by SOR. To address this, a lightweight channel attention mechanism is applied. It computes adaptive channel weights by leveraging both global average pooling $G_{\mathrm{avg}}(\cdot)$ and global max pooling $G_{\mathrm{max}}(\cdot)$, formulated as

$$\mathbf{s}_{\mathrm{HS}} = \mathrm{Si}\left(W_2\,\mathrm{Re}\left(W_1\left[G_{\mathrm{avg}}\left(\tilde{\mathbf{F}}_{\mathrm{HS}}\right) \parallel G_{\mathrm{max}}\left(\tilde{\mathbf{F}}_{\mathrm{HS}}\right)\right]\right)\right) \tag{4}$$

where $W_1 \in \mathbb{R}^{128\times(2C_\lambda)}$ and $W_2 \in \mathbb{R}^{C_\lambda\times 128}$ are learnable matrices, Si and Re are the sigmoid and ReLU functions. Then, $\mathbf{s}_{\mathrm{HS}}$ is reshaped to $1 \times 1 \times C_\lambda$ for generating $\mathbf{O}_{\mathrm{HS}} = \mathbf{s}_{\mathrm{HS}} \odot \tilde{\mathbf{F}}_{\mathrm{HS}}$, where $\odot$ is the elementwise multiplication.

3) G_{SAR}: G_{SAR} incorporates a deformable and multiscale orientation-enhanced SAR (DMOE-SAR) module with dual-pooling spatial attention. DMOE-SAR captures geometric patterns, directional structures, and other salient features through three parallel branches: a deformable sampling branch for adaptive geometric reception, an orientation enhancement branch to encode directional sensitivity, and a lightweight multiscale energy context enhancement branch. Given that SAR backscattering signals exhibit strong orientation dependence and are always affected by geometric distortions, DMOE-SAR considers learnable deformable convolutions and applies explicit orientation reinforcement at 0°, 45°, and 90° to better model these characteristics.

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a) Deformable sampling branch: The SAR input $\mathbf{X}_{\mathrm{SAR}} \in \mathbb{R}^{H\times W\times C_s}$ is processed by a 3×3 convolutional layer $\mathbf{F}_{\mathrm{off}}$, which predicts sampling offsets $\Delta p_r = (\Delta x, \Delta y)$ for each location r in the grid $\mathcal{R} = (-1,-1),(-1,0),\ldots,(1,1)$. Each offset corresponds to a spatial shift $(\Delta x, \Delta y)$. For a target location p, the deformable convolution is formulated as

$$y_{\mathrm{def}}(p) = \sum_{r\in\mathcal{R}} w(r)\cdot \mathcal{B}\left(\mathbf{X}_{\mathrm{SAR}}, p + r + \mathbf{F}_{\mathrm{off}}(\mathbf{X}_{\mathrm{SAR}})_r\right) \tag{5}$$

where $\mathcal{B}$ denotes the bilinear interpolation operation for handling fractional sampling positions. The branch output is denoted as $\mathbf{Z}_{\mathrm{def}} \in \mathbb{R}^{H\times W\times 64}$.

b) Orientation enhancement branch: SAR imagery is defined in the azimuth–range coordinate system, where azimuth and range form two physically meaningful orthogonal axes [35]. Thus, 0° and 90° are used as principal bases. Since the branch encodes **unsigned orientation** rather than **signed direction**, $\theta \equiv \theta + \pi$ and angles separated by 180° are redundant [36]. A single intermediate basis, i.e., 45°, is added to cover common oblique patterns. Denser angle sets require repeated interpolated rotations on discrete grids, increasing cost and potentially introducing discretization artifacts [37]. To enhance directional sensitivity, three parallel $k\times k$ convolutional branches with independent parameters are constructed. For an angle $\theta \in \{0, \pi/4, \pi/2\}$, the rotation matrix is defined as

$$\mathbf{R}_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}. \tag{6}$$

The rotated feature $\mathcal{T}_\theta(\cdot)$ is obtained by applying the inverse coordinate transformation, i.e., each pixel value at coordinate p is sampled from the original feature at position $\mathbf{R}_\theta^{-1}p$, with bilinear interpolation used for noninteger coordinates. The features $\mathbf{Z}_0 = \mathrm{Conv}_{3\times3}(\mathbf{X}_{\mathrm{SAR}})$, $\mathbf{Z}_{\pi/4} = \mathcal{T}_{\pi/4}(\mathrm{Conv}_{3\times3}(\mathbf{X}_{\mathrm{SAR}}))$, and $\mathbf{Z}_{\pi/2} = \mathcal{T}_{\pi/2}(\mathrm{Conv}_{3\times3}(\mathbf{X}_{\mathrm{SAR}}))$ are concatenated along the channel dimension for $\mathbf{Z}_{\mathrm{ori}} \in \mathbb{R}^{H\times W\times 48}$.

c) Multiscale energy context enhancement branch: To incorporate energy distribution context, 2×2 max pooling and 2×2 average pooling are applied to $\mathbf{X}_{\mathrm{SAR}}$ for downsampled features $\mathbf{P}_1$ and $\mathbf{P}_2$. Both features are processed by 1×1 convolutions at the downsampled resolution to maintain channel dimensions, then concatenated and upsampled by a transposed convolution. Let $\tilde{\mathbf{P}}_1$ and $\tilde{\mathbf{P}}_2$ denote the upsampled features at the original resolution. The obtained features are denoted as $\mathbf{Z}_{\mathrm{msc}} = [\tilde{\mathbf{P}}_1 \,\|\, \tilde{\mathbf{P}}_2] \in \mathbb{R}^{H\times W\times 64}$.

d) High-level feature fusion: The features $\mathbf{Z}_{\mathrm{def}}$, $\mathbf{Z}_{\mathrm{ori}}$, and $\mathbf{Z}_{\mathrm{msc}}$ are concatenated along channels for $\mathbf{F}_{\mathrm{SAR}} \in \mathbb{R}^{H\times W\times 176}$, which is then projected to C_s channels by a 3×3 transposed convolutional layer to yield $\tilde{\mathbf{F}}_{\mathrm{SAR}} \in \mathbb{R}^{H\times W\times C_s}$. Since SAR data exhibit a high sensitivity to local salient regions, a spatial attention mechanism is applied to $\tilde{\mathbf{F}}_{\mathrm{SAR}}$ to amplify informative spatial responses. Similar to (4), a spatial attention map $\mathbf{s}_{\mathrm{SAR}}$ is obtained. The output is $\mathbf{O}_{\mathrm{SAR}} = \mathbf{s}_{\mathrm{SAR}} \odot \tilde{\mathbf{F}}_{\mathrm{SAR}}$.

B. Discriminators in MDGAN

The discriminator structure in MDGAN is shown in Fig. 3. The discriminator distinguishes real samples from generated samples and provides gradient signals for generator optimization. To overcome the limitations of conventional CNN-based discriminators, a residual separable U-Net discriminator called RSU-NetD is constructed. It incorporates DSCA ResBlocks in both the deep feature and skip-merge stages (where skip tensors are resized if needed) to reinforce spectral–spatial feature representation. While the U-Net structure benefits from feature reuse by skip connections and residual pathways, it is susceptible to feature coadaptation and overfitting. To mitigate this, a layer-channel adaptive DropPath mechanism is applied to each residual branch, for which the drop rate is adjusted based on channel importance scores. This regularizer reduces feature redundancy and stabilizes gradients, improving robustness and generalization. The inputs to the DSCA ResBlock are $\mathbf{O}_{\mathrm{HS}}$ and $\mathbf{O}_{\mathrm{SAR}}$ for D_{HS} and D_{SAR}, respectively. D_{HS} and D_{SAR} share the same architecture but do not share parameters.

Take D_{HS} as an example: D_{HS} first generates latent features by some layers (as shown in Fig. 3), which are delivered into DSCA ResBlocks. In a DSCA ResBlock, the input is assumed as $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, where C is the channel number, and processed by two sequential 3×3 depthwise (DW) and 1×1 pointwise (PW) separable convolutions for $\mathbf{U}_1$ and $\mathbf{U}_2$. Instance normalization is used, followed by the leaky ReLU function for activation. Similar to (4), a channel attention map $\mathbf{s}_{\mathrm{CH}}$ is obtained for $\mathbf{s}_{\mathrm{CH}} \odot \mathbf{U}_2$. To suppress redundant feature responses and stabilize gradient propagation, $\mathbf{s}_{\mathrm{CH}} \odot \mathbf{U}_2$ is further processed by the layer-channel adaptive DropPath and added to $\mathbf{X}$ for information preservation.

In the lth DSCA ResBlock, the layer-channel adaptive DropPath input is simplified as $\mathbf{X}$ for a clear description. The drop probability $p_{l,c}$ for channel c is formulated as

$$p_{l,c} = \mathrm{Si}\left(\alpha l/L + \beta s_c\right), \quad c = 1,2,\ldots,C \tag{7}$$

where α and β are coefficients, L is the total block number, and s_c is the importance score of channel c, which is the cth value in $\mathbf{s}_{\mathrm{CH}}$. A binary mask is sampled from a Bernoulli distribution for each (b,h,w,c) to implement. The output $\widehat{\mathbf{X}}_{b,h,w,c}$ is obtained as

$$\begin{aligned} \widehat{\mathbf{X}}_{b,h,w,c} &= \frac{M_{b,h,w,c}}{1-p_{l,c}}\mathbf{X}_{b,h,w,c} \\ M_{b,h,w,c} &\sim \mathrm{Bernoulli}\left(1-p_{l,c}\right) \end{aligned} \tag{8}$$

where $M_{b,h,w,c} \in \{0,1\}$ is the elementwise mask indicating dropped (0) and retained (1) activations. Notably, $1/(1-p_{l,c})$ preserves the expected activation magnitude in training, preventing potential bias. Through stabilizing gradient propagation and suppressing noisy features, this regularization proves highly effective in deep networks with skip connections.

C. Transformer-Based Dual-Path Cross-Attention

The TDCA structure is shown in Fig. 4. To effectively align and interact HS and SAR features, TDCA with two inputs $\mathbf{O}_{\mathrm{HS}}$ and $\mathbf{O}_{\mathrm{SAR}}$ is constructed for complementary fusion. Both inputs are first projected into a compact latent space by 1×1 convolutions to generate query, key, and value representations in $\mathbb{R}^{H\times W\times 32}$, denoted as $(Q_{\mathrm{HS}}, K_{\mathrm{HS}}, V_{\mathrm{HS}})$ and $(Q_{\mathrm{SAR}}, K_{\mathrm{SAR}}, V_{\mathrm{SAR}})$. To enhance spatial discriminability, both keys are further enhanced by a 3×3 convolutional layer for $\widetilde{K}_{\mathrm{HS}}$ and $\widetilde{K}_{\mathrm{SAR}}$, respectively. The queries are then enhanced

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by conditioning them on the corresponding refined keys. In particular, Q_{HS} and $\widetilde{K}_{\mathrm{HS}}$ (Q_{SAR} and $\widetilde{K}_{\mathrm{SAR}}$) are concatenated and processed by two sequential 1×1 convolutions for the enhanced queries $\overline{Q}_{\mathrm{HS}}$ and $\overline{Q}_{\mathrm{SAR}}$.

Bidirectional cross-attention in the latent space is considered to exchange complementary information between HS and SAR features. The generated features are computed as

$$\mathbf{Z}_{\mathrm{SAR}\to\mathrm{HS}} = \mathrm{S}\left(\frac{\mathrm{F}\left(\overline{Q}_{\mathrm{HS}}\right)\ \mathrm{F}\left(\widetilde{K}_{\mathrm{SAR}}\right)^{\top}}{\sqrt{d}}\right)\mathrm{F}\left(V_{\mathrm{SAR}}\right) \tag{9}$$

$$\mathbf{Z}_{\mathrm{HS}\to\mathrm{SAR}} = \mathrm{S}\left(\frac{\mathrm{F}\left(\overline{Q}_{\mathrm{SAR}}\right)\ \mathrm{F}\left(\widetilde{K}_{\mathrm{HS}}\right)^{\top}}{\sqrt{d}}\right)\mathrm{F}\left(V_{\mathrm{HS}}\right) \tag{10}$$

where S is the softmax function, F(·) reshapes its input shape from $H \times W \times d$ to $HW \times d$. Both outputs are reshaped back to $H \times W \times d$ and projected to the original space by 1×1 convolutions. The final outputs are $\mathbf{X}'_{\mathrm{HS}} \in \mathbb{R}^{H\times W\times C_\lambda}$ and $\mathbf{X}'_{\mathrm{SAR}} \in \mathbb{R}^{H\times W\times C_s}$, which are mapped into a shared fusion space with C_F channels as $H_{\mathrm{proj}} = \mathrm{Conv}_{1\times1}(\mathbf{X}'_{\mathrm{HS}})$ and $S_{\mathrm{proj}} = \mathrm{Conv}_{1\times1}(\mathbf{X}'_{\mathrm{SAR}})$. A gating mask is obtained as

$$\mathcal{G} = \mathrm{Si}\left(\mathrm{LN}\left(\mathrm{Conv}_{1\times1}\left(\left[\mathbf{X}'_{\mathrm{HS}} \parallel \mathbf{X}'_{\mathrm{SAR}}\right]\right)\right)\right) \tag{11}$$

where $\mathcal{G} \in [0,1]^{H\times W\times C_F}$, LN(·) is layer normalization, Si is the sigmoid function. Then, H_{proj} and S_{proj} are fused by complementary weighting as

$$\mathbf{X}_F = \mathcal{G} \odot H_{\mathrm{proj}} + (1 - \mathcal{G}) \odot S_{\mathrm{proj}}. \tag{12}$$

The fused feature, processed by a 3×3 convolutional layer followed by layer normalization and global average pooling, is fed into a fully connected layer activated by the softmax function to produce classification results.

D. Network Optimization

The loss function of GenMSLnet is formulated as $\mathcal{L}_{\mathrm{total}} = \{\mathcal{L}_{\mathrm{G}}, \mathcal{L}_{\mathrm{D}}\}$, where $\mathcal{L}_{\mathrm{G}} = \lambda_h \mathcal{L}_G^h + \lambda_s \mathcal{L}_G^s + \lambda_c \mathcal{L}_{\mathrm{cls}} + \mathcal{L}_{\mathrm{SOR}}$, $\mathcal{L}_{\mathrm{D}} = \mathcal{L}_D^h + \mathcal{L}_D^s$, λ_h, λ_s, and λ_c are parameters. The generator losses $\mathcal{L}_G^h$ and $\mathcal{L}_G^s$ for HSI and SAR data are formulated as

$$\mathcal{L}_G^h = \mathbb{E}\left[-\log D_{\mathrm{HS}}\left(\mathbf{O}_{\mathrm{HS}}, \mathbf{X}_{\mathrm{HS}}\right)\right] \tag{13}$$

$$\mathcal{L}_G^s = \mathbb{E}\left[-\log D_{\mathrm{SAR}}\left(\mathbf{O}_{\mathrm{SAR}}, \mathbf{X}_{\mathrm{SAR}}\right)\right]. \tag{14}$$

The classification loss $\mathcal{L}_{\mathrm{cls}}$ is based on categorical cross-entropy. The SOR loss $\mathcal{L}_{\mathrm{SOR}}$ is as

$$\mathcal{L}_{\mathrm{SOR}} = \lambda_{\mathrm{diag}}\mathcal{R}_{\mathrm{diag}}\left(W^\ell\right) + \lambda_{\mathrm{off}}\mathcal{R}_{\mathrm{off}}\left(W^\ell\right) \tag{15}$$

$$\mathcal{R}_{\mathrm{diag}}\left(W^\ell\right) = \left\|\mathrm{diag}\left(W^\ell\left(W^\ell\right)^{\top}\right) - \mathbf{1}\right\|_2^2 \tag{16}$$

$$\mathcal{R}_{\mathrm{off}}\left(W^\ell\right) = \left\|\left(W^\ell\left(W^\ell\right)^{\top}\right) \odot (\mathbf{1} - I)\right\|_1 \tag{17}$$

where W^ℓ is the unfolded 2-D weight matrix of the ℓth convolutional layer, $\|\cdot\|_1$ and $\|\cdot\|_2$ are the ℓ_1 and ℓ_2 norms, respectively. Deviations of diagonal elements from 1 and large off-diagonal entries are penalized. The discriminator losses $\mathcal{L}_D^h$ and $\mathcal{L}_D^s$ for HSI and SAR data are formulated as $\mathcal{L}_D^h = \mathbb{E}[-\log D_{\mathrm{HS}}(\mathbf{X}_{\mathrm{HS}}, \mathbf{X}_{\mathrm{HS}})] + \mathbb{E}[-\log(1 - D_{\mathrm{HS}}(\mathbf{O}_{\mathrm{HS}}, \mathbf{X}_{\mathrm{HS}}))]$ and $\mathcal{L}_D^s = \mathbb{E}[-\log D_{\mathrm{SAR}}(\mathbf{X}_{\mathrm{SAR}}, \mathbf{X}_{\mathrm{SAR}})] + \mathbb{E}[-\log(1 - D_{\mathrm{SAR}}(\mathbf{O}_{\mathrm{SAR}}, \mathbf{X}_{\mathrm{SAR}}))]$. GenMSLnet alternatively minimizes $\mathcal{L}_{\mathrm{G}}$ and $\mathcal{L}_{\mathrm{D}}$ to optimize the generators and discriminators for robust representations for classification.

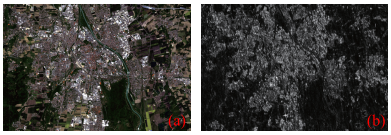


Fig. 5. Visualization of the Augsburg dataset, including (a) HSI with pseudocolor and (b) SAR data.

TABLE II
SAMPLE NUMBERS IN THE AUGSBURG DATASET, RA IS RESIDENTIAL AREA, IA IS INDUSTRIAL AREA, AND CA IS COMMERCIAL AREA

Category	land cover	Training	Testing	Total
1	Forest	146	13361	13507
2	RA	264	30065	30329
3	IA	21	3830	3851
4	Low Plants	248	26609	26857
5	Allotment	52	523	575
6	CA	7	1638	1645
7	Water	23	1507	1530
Total	/	761	77533	78294

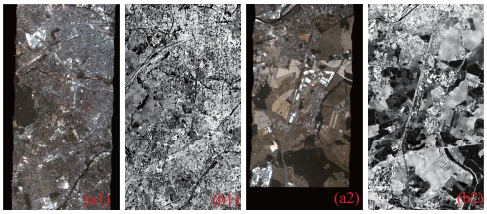


Fig. 6. Visualization of the (*1) Berlin-1 and (*2) Berlin-2 datasets, including (a1) and (a2) HSI with pseudocolor and the (b1) and (b2) SAR data.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset Description

The Augsburg dataset was acquired by the HySpex sensor over Augsburg, Germany. It comprises an HSI covering 180 spectral bands in the 400–2500-nm range and a dual-polarization SAR data spanning four channels. Both modalities have 332×485 pixels and a spatial resolution of 30 m. The sample numbers in all categories are provided in Table II. The visualization of HSI and SAR data is shown in Fig. 5.

The Berlin dataset is a typical HSI and SAR data fusion benchmark, which was acquired over both urban and rural areas in Berlin, Germany. Its HSI is an EnMAP-like scene simulated from HyMap data with 797×220 samples, a 30-m ground sampling distance (GSD), and 244 spectral bands from 400 to 2500 nm. Its SAR data were derived from Sentinel-1 dual-polarization single-look complex products processed by the ESA SNAP toolbox, yielding 1723×476 samples at 13.89-m GSD with four polarimetric features. For simplification, this dataset was split into two subscenes, i.e., **Berlin-1** and **Berlin-2**. The sample numbers in all categories of both subscenes are provided in Table III. The HSI and SAR data visualizations of both subscenes are shown in Fig. 6.

The sample ratio is the ratio of sample numbers in single categories relative to the total sample numbers. The categories with less than 2‰ sample ratios were removed. In particular, 1% of the samples in each category were randomly chosen for training. The remaining 99% of the samples were used for testing.

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TABLE III
SAMPLE NUMBERS IN THE BERLIN-1 AND BERLIN-2 DATASETS

Category	land cover	Berlin-1			Berlin-2		
		Training	Testing	Total	Training	Testing	Total
1	Forest	33	3329	3362	515	51077	51592
2	RA	2192	217063	219255	493	48894	49387
3	IA	23	2333	2356	172	17038	17210
4	Low Plants	86	8557	8643	592	58690	59282
5	Soil	194	19262	19456	174	17252	17426
6	Allotment	50	5024	5074	46	4616	4662
7	CA	/	/	/	53	5315	5368
8	Water	/	/	/	15	1583	1598
Total	/	2578	255568	258146	2060	204465	206525

B. Experimental Setting

All experiments were conducted on a CPU server without GPU acceleration. The server was equipped with two CPUs, each providing 188 GB of memory as indicated in the platform specifications. The code was implemented using the Keras framework. The adaptive moment estimation (Adam) was chosen. In SOR, the tradeoff parameters λ_1 and λ_2 were set as 0.005 and 0.0005, respectively. The coefficients α and β were set as 1.0 and 0.5, respectively. The parameters λ_h, λ_s, and λ_c were set as 1. Each training epoch consists of ten sequential training steps for the generator and discriminator. The learning rate was set as 5×10^{-3}. The batch size was selected within $\{64, 128, 256, 512\}$. Before delivering both RS modalities into networks, an $H \times W$ sliding window partitions the modalities into patches $\mathbf{X}_{\mathrm{HS}} \in \mathbb{R}^{H \times W \times C_\lambda}$ and $\mathbf{X}_{\mathrm{SAR}} \in \mathbb{R}^{H \times W \times C_s}$. For simplification, H and W were set as the same value, chosen within $\{3, 5, 7, 9, 11\}$. All experiments were repeated ten times to mitigate the effect of random sampling bias. The training process was terminated when all network parameters converged. Six evaluation metrics were selected, including the accuracy per class, overall accuracy (OA), average accuracy (AA), precision (PR), Kappa coefficient (KC), and $F1$-score ($F1$). Twenty state-of-the-art approaches were selected, including three fusion baselines (i.e., early fusion, middle fusion, and late fusion) [38], ENDnet [38], two CNN-based networks, i.e., CNN_H and CNN_L [39], enhanced multisource fusion network (EMFNet) [40], multiview hierarchical network (MVHN) [41], multiscale self-calibrated convolution network (MSNetSC) [42], local information interaction Transformer (LIIT) [43], morphological convolution and attention calibration network (MCAC) [44], hashing-based deep metric learning (HDML) method [45], coupled patch CNN (CPCNN) [39], dual CNN (DualCNN) [46], generative adversarial fusion network (FGAN) [47], dual discriminator conditional GAN (DDcGAN) [48], RESTORE-DiT [49], DDPM [50], HyperPointFormer [51], and FusDreamer [52]. Their descriptions are as follows.

1) *ENDnet* reconstructs its inputs using fully connected layers and ignores spatial information.
2) *CNN_H* and *CNN_L* use similar architectures, using stacked convolutional layers followed by a fully connected layer activated by the softmax function.
3) *EMFnet* adopts an encoder–decoder with attention-guided cross-connection blocks and recalibration for fusion.
4) *MVHN* uses principal component analysis with Gabor filtering for feature extraction, and leverages a residual fusion block to integrate HSI and SAR data.
5) *MSNetSC* uses a hierarchical residual structure with self-calibrated convolutions and multiple self-attention mechanisms to improve model performance.
6) *LIIT* uses a transformer based on multihead self-attention to enhance local spatial and spectral interactions.
7) *MCAC* combines morphological convolution and attention alignment mechanisms for joint classification.
8) *HDML* uses attention-guided fusion modules to extract discriminative joint representations, capturing both intramodal and cross-modal feature dependencies.
9) *CPCNN* adopts coupled patch-based CNNs to optimize HS and SAR branches under a unified loss, enabling fine-grained complementary feature extraction.
10) *DualCNN* comprises parallel branches dedicated to learning spectral and spatial representations separately, which are concatenated for HS and SAR joint classification.
11) *FGAN* uses a generator–discriminator framework to integrate complementary modalities simultaneously for optimizing adversarial training and structural similarity.
12) *DDcGAN* enhances the adversarial fusion approach by incorporating dual conditional discriminators, ensuring multiresolution consistency across RS modalities.
13) *RESTORE-DiT* (RES-DiT for short) proposes a conditional diffusion framework on a sequential denoising Transformer, guiding the iterative denoising of cloud-contaminated optical time series for reconstruction.
14) *DDPM* introduces a diffusion-based fusion paradigm for RS image fusion, using iterative denoising to learn robust multimodal representations to benefit downstream classification.
15) *HyperPointFormer* (HPF for short) performs 3-D multimodal fusion using dual-branch cross-attention Transformers, capturing cross-modal interactions to enhance performance.
16) *FusDreamer* (FusD for short) presents a label-efficient multimodal RS world model, leveraging diffusion-based generative modeling to learn transferable representations and improve classification under limited annotations.

C. Experimental Results and Comparison

1) Quantitative Comparison: Classification accuracies and their standard deviations over ten repeated runs on three datasets are provided in Tables IV–VI, where the best accuracies are marked in bold. The reported standard deviations indicate that GenMSLnet maintains competitive stability across repeated runs while achieving the best average performance on these main evaluation metrics. For the Augsburg dataset, early and late fusion achieve nearly identical OAs, indicating that simple concatenation cannot surpass the performance ceiling. This limitation arises primarily because the dataset contains complex boundaries and rich textures, requiring not only spectral features but also local spatial structures and cross-modal textural information for accurate recognition. ENDnet improves spectral consistency through reconstruction, but still underperforms on complex categories due to insufficient modeling of multiscale textures and spatial structures. Single-modal baselines CNN_H and CNN_L perform worse than

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TABLE IV

ACCURACIES WITH STANDARD DEVIATIONS, INCLUDING OA (%), AA (%), PR (%), KC (×100), AND $F1$ (×100), ON THE AUGSBURG DATASET

Class	Early	Middle	Late	ENDnet	CNN_H	CNN_L	EMFnet	MVHN	MSNetSC	LIIT	MCAC	HDML	CPCNN	DualCNN	FGAN	DDcGAN	RES-DiT	DDPM	HPF	FusD	Our
1	94.71	96.14	95.02	93.92	97.28	93.05	88.14	98.02	93.13	95.51	90.58	92.89	92.41	97.07	97.50	**98.26**	88.37	97.86	85.60	93.75	97.01
2	95.03	88.97	93.66	94.58	92.55	94.70	97.60	95.70	97.81	96.92	95.48	97.78	**98.72**	98.44	98.54	98.08	95.39	97.17	96.39	94.05	96.77
3	26.32	31.85	34.05	17.52	47.47	0	28.80	48.64	32.14	53.76	8.30	19.40	16.81	49.82	34.39	49.66	47.73	44.07	44.20	**81.93**	55.54
4	93.99	88.26	93.75	93.34	86.08	91.34	94.77	88.58	94.81	93.23	91.12	95.24	96.53	96.97	**98.14**	96.69	94.83	97.90	95.61	96.02	97.67
5	61.95	55.64	46.27	42.45	57.55	7.27	32.12	52.96	51.05	**67.69**	43.79	49.71	51.43	49.71	50.10	54.11	30.59	44.55	35.95	55.45	57.36
6	2.81	5.31	2.75	3.48	1.47	0	12.39	1.65	1.65	0.55	1.16	1.59	0.43	3.11	0.49	2.93	33.52	21.18	**37.73**	4.82	21.18
7	29.33	46.12	47.98	45.99	33.51	0.40	42.14	47.84	46.58	**48.37**	40.15	29.06	47.25	39.48	45.52	45.72	30.66	20.17	44.06	29.40	45.32
OA	87.78	84.32	87.85	87.01	85.61	84.16	88.28	88.13	89.39	90.10	85.42	88.50	89.44	90.98	91.62	91.92	88.63	91.46	89.02	90.67	**92.22**
OA_{STD}	±0.21	±0.28	±0.25	±0.33	±0.16	±0.13	±0.50	±0.84	±0.46	±0.43	±0.23	±0.92	±0.23	±0.28	±1.08	±1.46	±1.14	±0.59	±0.30	±0.86	±0.77
AA	57.53	58.90	59.07	55.90	59.42	40.97	56.57	61.91	59.60	65.15	52.94	55.10	57.65	61.51	60.67	63.64	60.16	60.42	62.79	65.06	**67.26**
AA_{STD}	±0.31	±0.32	±0.38	±0.55	±0.20	±0.13	±0.75	±1.09	±0.94	±1.15	±0.34	±1.39	±0.44	±0.31	±1.78	±2.37	±1.95	±0.85	±0.52	±1.64	±1.51
PR	60.38	56.30	58.08	59.29	61.53	38.77	63.12	62.33	65.79	65.11	58.30	66.30	67.03	68.91	71.42	68.18	63.90	70.15	65.75	67.76	**73.36**
PR_{STD}	±0.33	±0.31	±0.37	±0.58	±0.21	±0.12	±0.84	±1.10	±1.04	±1.15	±0.37	±1.67	±0.51	±0.35	±2.10	±2.54	±2.07	±0.99	±0.54	±1.71	±1.65
KC	82.24	77.77	82.46	81.11	79.32	76.58	82.85	82.86	84.48	85.71	78.63	83.03	84.38	87.83	87.70	88.24	83.65	87.61	84.16	86.66	**88.76**
KC_{STD}	±0.44	±0.42	±0.53	±0.80	±0.27	±0.24	±1.10	±1.54	±1.33	±1.51	±0.51	±2.10	±0.64	±0.44	±2.57	±3.29	±2.71	±1.23	±0.70	±2.19	±1.99
F1	54.83	54.77	56.56	54.31	56.55	38.92	57.54	59.89	58.73	61.65	50.46	54.90	57.66	63.01	62.01	63.61	60.27	61.41	62.83	61.96	**68.49**
$F1_{STD}$	±0.30	±0.30	±0.36	±0.53	±0.19	±0.12	±0.76	±1.05	±0.93	±1.09	±0.32	±1.39	±0.44	±0.32	±1.82	±2.37	±1.95	±0.86	±0.52	±1.56	±1.54

TABLE V

ACCURACIES WITH STANDARD DEVIATIONS, INCLUDING OA (%), AA (%), PR (%), KC (×100), AND $F1$ (×100), ON THE BERLIN-1 DATASET

Class	Early	Middle	Late	ENDnet	CNN_H	CNN_L	EMFnet	MVHN	MSNetSC	LIIT	MCAC	HDML	CPCNN	DualCNN	FGAN	DDcGAN	RES-DiT	DDPM	HPF	FusD	Our
1	9.21	8.79	9.74	5.76	33.47	0	27.72	13.59	25.71	16.32	0	0.03	**45.44**	37.18	23.46	26.34	10.17	37.23	41.86	36.69	45.09
2	88.88	90.46	90.59	92.61	93.37	98.99	97.73	97.45	97.47	98.12	**99.46**	97.86	98.30	97.22	98.07	97.77	98.61	98.73	97.64	98.21	97.73
3	2.36	8.92	3.09	1.07	2.40	0	6.45	1.11	4.21	2.96	0.04	0	4.50	**21.59**	0.04	0.26	6.52	2.57	15.33	1.63	0.56
4	41.67	18.98	28.57	13.97	53.39	15.98	47.01	31.63	37.23	43.16	0.04	31.51	48.86	51.95	41.25	43.76	37.26	45.49	31.29	36.90	**59.07**
5	29.77	27.68	42.49	30.50	**45.87**	0.89	27.39	35.23	32.29	29.13	1.60	30.54	34.43	37.54	30.64	34.97	33.05	34.03	42.60	28.19	41.91
6	53.16	35.67	46.66	45.25	46.32	37.53	49.18	50.22	54.08	45.67	36.76	36.62	60.52	66.87	51.70	48.13	64.16	46.63	71.76	63.52	**75.72**
OA	80.31	80.46	82.18	82.41	85.92	85.42	88.03	87.65	87.91	88.11	85.31	87.19	89.54	89.14	88.31	88.43	88.95	89.37	89.29	88.51	**90.22**
OA_{STD}	±0.43	±0.63	±0.60	±0.55	±0.31	±0.34	±0.85	±1.03	±0.94	±0.95	±0.35	±1.14	±0.50	±0.44	±1.21	±1.86	±1.28	±1.14	±0.83	±1.26	±1.15
AA	37.51	31.75	36.86	31.53	45.80	25.56	42.58	38.21	41.83	39.23	22.98	32.76	48.68	52.06	40.86	41.87	41.63	44.11	50.08	44.19	**53.35**
AA_{STD}	±0.79	±0.90	±0.87	±1.04	±0.80	±0.70	±1.03	±1.18	±1.09	±1.33	±0.85	±1.52	±0.99	±0.70	±1.96	±3.48	±2.12	±1.40	±1.19	±1.58	±1.53
PR	34.93	32.94	40.40	37.94	52.31	36.76	67.26	63.68	63.90	73.79	32.21	54.88	80.28	66.50	81.18	68.93	73.73	**86.44**	67.65	63.54	82.13
PR_{STD}	±0.72	±0.85	±0.82	±0.94	±0.70	±0.63	±0.99	±1.15	±1.06	±1.25	±0.75	±1.44	±0.89	±0.65	±1.81	±3.16	±1.95	±1.35	±1.12	±1.52	±1.45
KC	29.58	24.63	32.69	25.81	45.09	15.65	43.20	41.48	43.23	42.08	9.69	36.09	51.12	52.70	43.51	45.50	46.57	48.41	52.30	45.01	**57.23**
KC_{STD}	±0.54	±0.71	±0.68	±0.70	±0.46	±0.45	±0.90	±1.08	±0.99	±1.06	±0.50	±1.25	±0.65	±0.52	±1.44	±2.35	±1.53	±1.22	±0.94	±1.36	±1.26
F1	35.19	31.94	38.03	33.24	47.82	27.26	49.85	44.74	48.77	46.72	24.05	38.21	56.02	57.54	48.10	49.17	48.61	52.99	56.20	50.24	**58.34**
$F1_{STD}$	±0.75	±0.87	±0.84	±0.99	±0.75	±0.66	±1.01	±1.17	±1.08	±1.29	±0.80	±1.48	±0.94	±0.67	±1.89	±3.32	±2.04	±1.37	±1.15	±1.55	±1.49

TABLE VI

ACCURACIES WITH STANDARD DEVIATIONS, INCLUDING OA (%), AA (%), PR (%), KC (×100), AND $F1$ (×100), ON THE BERLIN-2 DATASET

Class	Early	Middle	Late	ENDnet	CNN_H	CNN_L	EMFnet	MVHN	MSNetSC	LIIT	MCAC	HDML	CPCNN	DualCNN	FGAN	DDcGAN	RES-DiT	DDPM	HPF	FusD	Our
1	86.83	89.25	86.80	85.75	91.01	69.12	89.59	91.89	89.49	90.75	86.35	88.92	89.71	89.77	91.55	**93.45**	87.67	90.63	90.43	90.11	92.68
2	82.40	73.95	80.00	78.85	82.12	72.85	85.12	79.84	84.89	87.04	77.33	84.99	84.37	85.22	85.28	85.96	82.68	**89.25**	81.62	82.38	85.92
3	63.03	59.94	63.82	67.36	72.51	42.53	71.98	69.24	72.81	66.41	49.48	65.33	71.73	74.23	74.42	75.28	64.52	74.47	73.88	57.67	**77.56**
4	86.81	82.46	85.09	84.38	88.47	71.55	90.79	85.86	88.95	90.25	85.61	88.26	**92.46**	90.25	90.29	91.25	89.29	91.80	89.81	86.85	92.08
5	81.75	85.10	83.26	80.43	88.91	49.98	83.90	88.05	**92.03**	91.27	78.54	85.98	85.56	86.96	91.11	89.35	87.77	90.28	81.89	84.04	89.40
6	7.13	12.00	8.60	4.62	20.17	0.07	27.80	21.32	21.44	33.49	8.62	4.94	21.19	25.58	32.50	27.59	4.67	18.01	17.23	23.37	**47.08**
7	2.67	7.29	4.24	6.68	5.38	0.06	13.66	12.21	20.60	**21.09**	6.17	1.37	15.46	12.23	13.03	13.29	14.08	7.69	4.61	2.99	4.65
8	16.28	12.75	13.29	20.90	0.06	0.88	61.35	46.36	61.24	62.54	8.45	15.58	51.14	44.92	58.69	33.27	79.89	26.65	77.09	63.15	**83.47**
OA	78.82	76.38	77.98	77.34	81.91	62.99	83.33	81.05	83.51	84.42	75.81	80.83	83.60	83.48	84.60	85.13	81.17	84.97	82.06	81.20	**85.97**
OA_{STD}	±0.50	±0.61	±0.61	±0.72	±0.68	±0.77	±0.95	±0.84	±0.94	±0.92	±0.49	±1.06	±0.52	±0.46	±1.09	±1.64	±1.21	±0.99	±0.79	±1.14	±1.03
AA	53.36	52.84	53.14	53.62	56.08	38.38	65.53	61.85	66.43	67.86	50.07	54.42	63.95	63.65	67.11	63.68	63.82	61.10	64.57	62.47	**71.60**
AA_{STD}	±0.62	±0.73	±0.73	±1.05	±0.64	±0.55	±1.01	±1.48	±0.94	±0.93	±0.68	±1.53	±0.85	±0.77	±1.69	±2.12	±2.46	±1.12	±0.83	±2.26	±1.59
PR	60.64	56.60	56.28	56.35	71.96	52.08	74.77	68.90	71.56	76.19	55.70	71.08	77.98	75.15	76.27	78.59	64.52	**78.92**	70.45	68.57	76.25
PR_{STD}	±0.58	±0.69	±0.69	±0.95	±0.65	±0.62	±0.99	±1.29	±0.94	±0.93	±0.62	±1.39	±0.75	±0.68	±1.51	±1.98	±2.09	±1.08	±0.82	±1.92	±1.39
KC	72.59	69.64	71.64	70.83	76.62	51.46	78.47	75.62	78.84	79.90	68.75	75.12	78.77	78.68	80.15	80.78	75.70	80.54	76.80	74.90	**81.93**
KC_{STD}	±0.30	±0.35	±0.35	±0.49	±0.32	±0.30	±0.50	±0.68	±0.47	±0.46	±0.32	±0.72	±0.39	±0.35	±0.79	±1.01	±1.11	±0.55	±0.41	±1.02	±0.73
F1	54.48	53.43	53.55	54.13	56.46	38.69	68.52	64.09	68.21	70.96	51.11	55.46	67.84	66.83	69.80	67.25	62.14	64.16	65.53	63.28	**72.73**
$F1_{STD}$	±0.60	±0.71	±0.71	±0.98	±0.65	±0.59	±1.00	±1.35	±0.94	±0.93	±0.64	±1.44	±0.78	±0.71	±1.57	±2.02	±2.21	±1.09	±0.82	±2.04	±1.46

GenMSLnet, indicating that a single modality is insufficient to balance global consistency and local details under severe spectral–spatial mixing. Generative-based methods, including FGAN, DDcGAN, DDPM, and FusD, generally outperform traditional fusion baselines, suggesting that generation strategies can mitigate limited sample challenges. However, these methods struggle to balance performance on hard minority classes with overall generalization. While FusD achieves high accuracy on *Industrial Area* with 21 training samples, its OA remains 90.67%, suggesting that improving only a few challenging categories is insufficient for global optimization. By contrast, GenMSLnet achieves the best performance with an OA of 92.22% ± 0.77%, outperforming the second-best method DDcGAN (91.92% ± 1.46%) by 0.30%. Notably, it exhibits robust and satisfactory performance across balanced evaluation metrics, achieving an AA of 67.26% ± 1.51% and an $F1$ of 0.6849 ± 1.54. The consistent improvements reflect enhanced class balance and greater prediction stability. These results validate that modality-specific learning, when integrated with cross-modal interaction, mitigates representation bias and strengthens the capturing of spectral–spatial mixtures and structural textures.

On the Berlin-1 and Berlin-2 datasets, the results exhibit strong consistency. GenMSLnet achieves the highest OA of

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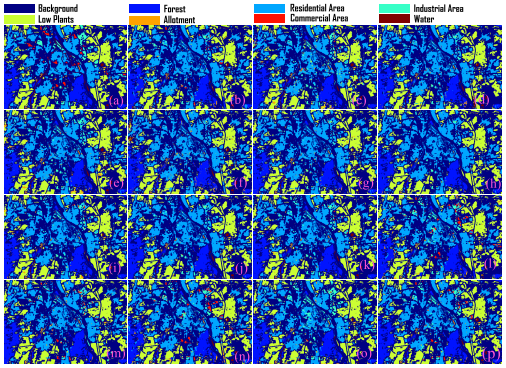


Fig. 7. (a) Ground truth and (b)–(p) classification maps on the Augsburg dataset, including (b) EMFnet, (c) MVHN, (d) MSNetSC, (e) LIIT, (f) MCAC, (g) HDML, (h) CPCNN, (i) DualCNN, (j) FGAN, (k) DDcGAN, (l) RES-DiT, (m) DDPM, (n) HPF, (o) FusD, and (p) GenMSLnet.

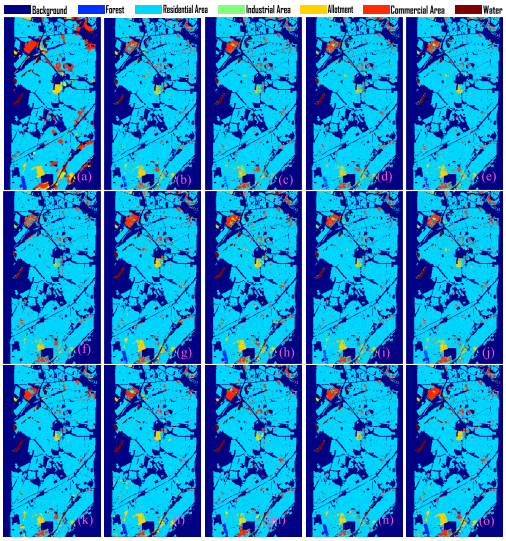


Fig. 8. (a) Ground truth and (b)–(o) classification maps on the Berlin-1 dataset, including (b) EMFnet, (c) MVHN, (d) MSNetSC, (e) LIIT, (f) HDML, (g) CPCNN, (h) DualCNN, (i) FGAN, (j) DDcGAN, (k) RES-DiT, (l) DDPM, (m) HPF, (n) FusD, and (o) GenMSLnet.

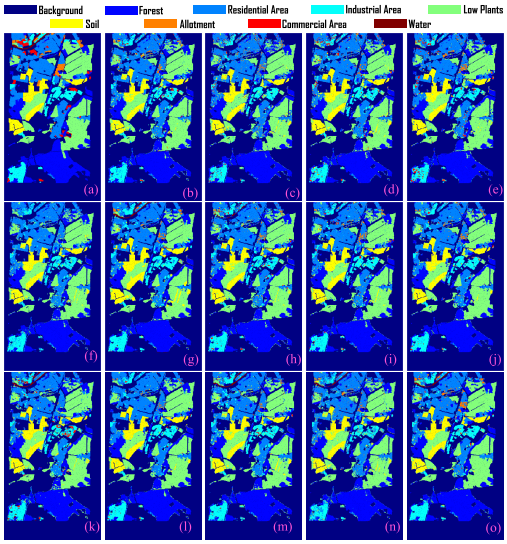


Fig. 9. (a) Ground truth and (b)–(o) classification maps on the Berlin-2 dataset, including (b) EMFnet, (c) MVHN, (d) MSNetSC, (e) LIIT, (f) HDML, (g) CPCNN, (h) DualCNN, (i) FGAN, (j) DDcGAN, (k) RES-DiT, (l) DDPM, (m) HPF, (n) FusD, and (o) GenMSLnet.

TABLE VII

EFFICIENCY COMPARISON ON THE AUGSBURG DATASET, INCLUDING #PARA (M), FLOPS, TRAINING TIME (1000 EPOCHS, MIN), INFERENCE TIME (MS/BATCH), LATENCY (MS/SAMPLE), THROUGHPUT (THPT., SAMPLES/S), AND MEMORY CONSUMPTION (CPU RAM, GB)

Model	#Para	FLOPs	Training	Inference	Latency	Thpt.	CPU RAM
Early	0.09	1.03	3.36	8.26	0.11	9385.42	38.3
Middle	0.31	0.61	3.54	7.32	0.09	10598.72	35.7
Late	0.30	0.60	3.16	7.58	0.10	10226.75	31.6
ENDnet	0.33	0.65	6.33	12.54	0.16	6181.43	25.7
CNN_H	0.07	3.24	3.33	8.84	0.11	8769.53	5.2
CNN_L	0.02	0.98	3.00	10.78	0.14	7189.92	7.0
EMFnet	1.47	73.33	5.61	30.18	0.39	2569.20	35.5
MVHN	4.09	204.18	9.22	12.94	0.17	5991.08	21.9
MSNetSC	0.86	43.97	7.08	26.76	0.35	2897.81	13.3
LIIT	1.79	73.55	9.59	54.80	0.71	1414.82	33.4
MCAC	0.02	0.58	4.96	12.39	0.16	6257.34	36.7
HDML	6.52	321.58	10.51	42.16	0.54	1839.14	31.5
CPCNN	0.15	14.42	4.22	15.36	0.20	5047.61	31.2
DualCNN	0.08	4.11	4.00	13.89	0.18	5582.04	30.7
FGAN	7.60	73.71	151.04	58.63	0.76	1322.43	10.5
DDcGAN	12.27	285.93	228.03	85.31	1.10	908.88	14.8
RES-DiT	8.01	113.37	52.14	570.35	7.36	135.94	25.9
DDPM	3.18	50.76	9.38	58.14	0.75	1333.65	35.9
HPF	0.42	14.31	12.68	99.99	1.29	775.40	29.5
FusD	10.92	32.11	10.36	54.11	0.70	1433.01	36.6
Our	6.00	59.87	22.76	14.14	0.18	5484.62	17.5

90.22% ± 1.15% (Berlin-1) and 85.97% ± 1.03% (Berlin-2), along with the best AA, KC, and $F1$. Although DDPM yields the best PR values on both datasets (86.44% and 78.92%), GenMSLnet maintains superior performance when considering all evaluation metrics together. This indicates that GenMSLnet offers a more balanced and robust performance. Classification maps on three datasets are shown in Figs. 7–9. Consistent with the quantitative results, the GenMSLnet maps are more spatially coherent, with less noise and clearer structural boundaries.

To further verify whether relatively small OA improvements are statistically meaningful, a two-sided Wilcoxon signed-rank test is performed on the OA results from ten repeated runs. This nonparametric paired test evaluates whether the performance difference between two different methods is centered around zero across runs consistently. A result is regarded as statistically significant when its p-value is $p < 0.05$. On the Augsburg dataset, the three strongest competitors, FGAN, DDPM, and DDcGAN, are selected. Their OA gaps to GenMSLnet are smaller than 1%. The obtained p-values are as follows: $p_{\text{Ours versus FGAN}} = 0.0195$, $p_{\text{Ours versus DDPM}} = 0.0244$, $p_{\text{Ours versus DDcGAN}} = 0.0478$. The improvements of GenMSLnet over all three baselines on this dataset are statistically significant, although $p_{\text{Ours versus DDcGAN}}$ is close to the significance threshold. Experiments are further conducted on the Berlin-1 and Berlin-2 datasets. Their p-values between GenMSLnet and DDcGAN are 0.0098 (Berlin-1) and 0.0187 (Berlin-2), which are also statistically significant. These results indicate that our superiority over DDcGAN is consistently supported across datasets rather than being caused by random variation.

2) Computational Efficiency Comparison: Table VII shows the comparison of computational efficiencies in terms of parameters (#Para), floating-point operations (FLOPs) per

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TABLE VIII

SYSTEMATIC LEAVE-ONE-OUT ABLATION ON MAJOR COMPONENTS OF GENMSLNET ON THE AUGSBURG DATASET

Model	OA(%)	AA(%)	PR(%)	KC(×100)	F1(×100)
Full model (GenMSLnet)	**92.22**	**67.26**	**73.36**	**88.76**	**68.49**
Replacing TDCA with concatenation	90.90	62.88	71.74	86.95	63.45
Replacing MDGAN with a non-adversarial feature extractor	91.39	64.95	71.62	87.33	65.70
Disabling SOR in the generators	91.93	65.73	71.85	88.27	64.31
Removing the orientation-enhanced branch from G_{SAR}	91.86	65.37	71.39	88.17	66.66
Replacing LCADropPath with standard dropout	92.13	66.18	70.75	88.61	65.81
Replacing RSU-NetD with a plain CNN discriminator	91.69	66.40	71.10	87.99	65.77

TABLE IX

CLASSIFICATION ACCURACIES OF VARIANTS, INCLUDING OA (%), AA (%), PR (%), KC (×100), AND $F1$ (×100), ON THE AUGSBURG DATASET

Model	OA	AA	PR	KC	F1	Model	OA	AA	PR	KC	F1
A_1	92.03	66.75	71.19	88.48	66.68	C_1	91.40	64.17	70.80	87.48	65.51
A_2	91.98	66.20	70.98	88.16	65.98	C_2	91.88	65.56	71.60	88.26	66.27
A_3	91.82	66.68	71.62	88.18	66.17	C_3	91.76	66.76	71.98	88.10	66.40
A_4	91.52	63.13	70.14	87.75	64.34	C_4	91.77	64.90	71.39	88.08	65.50
A_5	91.96	65.79	71.86	88.34	67.10	C_5	91.79	67.16	71.26	88.15	67.17
B_1	92.00	66.25	72.14	88.37	66.43	K_1	88.86	57.98	69.59	83.52	58.29
B_2	92.07	66.18	72.49	88.43	67.16	K_2	90.17	61.50	70.09	85.61	62.24
B_3	92.01	67.19	71.77	88.46	67.37	K_3	90.93	62.91	71.71	86.76	63.61
B_4	91.89	66.62	71.41	88.30	67.96	K_4	91.06	62.66	72.79	86.92	64.24
-	-	-	-	-	-	K_5 (Our)	**92.22**	**67.26**	**73.36**	**88.76**	**68.49**

sample, training and inference times, latency, throughput (Thpt.), and memory consumption (CPU RAM) on the Augsburg dataset. Overall, lightweight CNN baselines have relatively lower computational costs. Early, middle, and late fusion techniques use fewer than 0.31M parameters, require 3.16–3.54 min for training, achieve inference times of 7.32–8.26 ms, and reach throughput exceeding 9385 samples/s. On the contrary, CNN_H exhibits higher FLOPs, demonstrating that fewer parameters do not guarantee lower computational consumption. On the contrary, generation-based methods are substantially more expensive. FGAN with 7.60M parameters and DDcGAN with 12.27M parameters need 151.04 and 228.03 min for training, respectively, and their inference times rise to 58.63 and 85.31 ms, reducing throughput to 1322.43 and 908.88 samples/s. Although the training time for RES-DiT is 52.14 min, its inference and latency times reach 570.35 and 7.36 ms. Its throughput is 135.94 samples/s, making this method impractical for fast deployment. DDPM and FusD reduce inference time; however, they still require about 56 ms for inference. In terms of CPU RAM, GenMSLnet maintains a moderate memory footprint among the compared methods. GenMSLnet with 6.00M parameters (half that of DDcGAN) provides a better efficiency–effectiveness tradeoff. It reduces training time from 228.03 to 22.76 min. Moreover, its inference time and throughput are 14.14 ms and 5484.62 samples/s (about six times faster than DDcGAN and over forty times faster than RES-DiT), while its CPU RAM consumption remains moderate at 17.5 GB. These results show that GenMSLnet avoids heavy inference consumption while being moderate in memory usage and competitive in performance.

D. Ablation and Parameter Selection Experiments

For clarity and consistency, all ablation and parameter selection experiments were conducted on the Augsburg dataset.

1) Systematic Ablation on Major Components: To clearly quantify the individual contribution of each major component, a systematic leave-one-out ablation study is conducted on the Augsburg dataset. Starting from the full model, each variant removes or replaces exactly one major component while keeping the remaining architecture, training protocol, data split (1% training), and hyperparameters unchanged. Table VIII provides a high-level summary of the major-component contributions, while the fine-grained principle- and component-level variants, i.e., $A_*/B_*/C_*/K_*$, are reported in Table IX.

Table VIII shows that replacing TDCA with simple concatenation causes the largest degradation, reducing OA by 1.32% and $F1$ by 0.0504. This underscores the necessity of cross-attention-based interaction and gated fusion for effectively exploiting HS-SAR complementarity beyond naive feature stacking. Replacing MDGAN with a nonadversarial feature extractor reduces OA by 0.83% and $F1$ by 0.0279. In particular, this variant removes only the adversarial learning part, including the discriminators and generator shell, while retaining the internal multiscale modules as dual-branch feature extractors. This degradation highlights the contribution of adversarial learning to learning more robust representations for downstream fusion under limited supervision. Disabling SOR results in a modest OA drop of 0.29% but a substantial $F1$ decline of 0.0418, suggesting that SOR primarily improves class-balanced recognition rather than OA. Removing the orientation-enhanced branch from G_{SAR} decreases $F1$ by 0.0183, confirming the effectiveness of explicit orientation modeling for capturing direction-sensitive backscattering features. Replacing LCADropPath with standard dropout decreases $F1$ by 0.0268. Replacing RSU-NetD with a plain CNN discriminator decreases $F1$ by 0.0272. These results indicate that both the proposed stability-oriented regularization and discriminator design contribute to robust adversarial optimization.

2) Necessity of Principles and Components: Three explicit design principles are provided to show that this improvement does not stem solely from increased parameters.

Principle 1 (P1) Modality-Specific Fidelity:

HSI and SAR data exhibit distinct characteristics. Compared to enforcing early homogenization, modality-specific modeling preserves modality-inherent distributions and discriminative patterns. For both modalities, P1 is implemented by MDGAN with modality-specific generators G_{HS} and G_{SAR}.

Principle 2 (P2) Generalization and Training Stability:

Joint learning with limited labels combined with adversarial augmentation is challenged by overfitting and training instability. Thus, regularization and stabilization mechanisms are necessary to enhance robustness and prevent degenerate solutions. GenMSLnet follows P2 by SOR regularization in the generators and RSU-NetD with adaptive DropPath.

Principle 3 (P3) Cross-Modal Complementary Alignment:

Given that HSI and SAR data provide complementary information, effective fusion should semantically align them and selectively propagate informative signals, avoiding naive concatenation or uniform aggregation. In GenMSLnet, P3 is

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realized by TDCA, conducting bidirectional cross-attention and complementary gated fusion in a shared latent space.

1) *Principle-Level Alternative Models:*

$\boldsymbol{K_1}$ (Base network) preserves the basic dual-branch backbone and applies channelwise concatenation for fusion, followed by a classifier (no MDGAN and no TDCA).

$\boldsymbol{K_2}$ ($\boldsymbol{K_1}$+ P1) introduces P1 in K_1.

$\boldsymbol{K_3}$ ($\boldsymbol{K_1}$+ P1 + P2) introduces P1 and P2 in K_1.

$\boldsymbol{K_4}$ ($\boldsymbol{K_1}$+ P3) introduces P3 in K_1.

$\boldsymbol{K_5}$ ($\boldsymbol{K_1}$+ P1 + P2 + P3) is the presented GenMSLnet.

2) *Component-Level Alternative Models:*

a) P1 Related Variants:

$\boldsymbol{A_1}$ removes the channel attention mechanism from G_{HS}.

$\boldsymbol{A_2}$ removes both the channel and spatial attention mechanisms from G_{HS} and G_{SAR}, respectively.

$\boldsymbol{A_3}$ replaces the multiscale structures in G_{HS} with stacked convolutional layers with comparable depth.

$\boldsymbol{A_4}$ replaces the multiscale structures in both G_{HS} and G_{SAR} with stacked convolutional layers with comparable depth.

$\boldsymbol{A_5}$ replaces the deformable convolutions in G_{SAR} with standard convolutions.

b) P2 Related Variants:

$\boldsymbol{B_1}$ replaces SOR with spectral normalization (SN) in generators.

$\boldsymbol{B_2}$ replaces SOR with ℓ_2 weight decay in generators.

$\boldsymbol{B_3}$ removes the DSCA ResBlocks from RSU-NetD and keeps the remaining discriminator structure unchanged.

$\boldsymbol{B_4}$ removes the stochastic noise input from the generators and keeps only the conditional feature input.

c) P3 Related Variants:

$\boldsymbol{C_1}$ removes the bidirectional cross-attention from TDCA, i.e., no cross-modal attention exchange is performed.

$\boldsymbol{C_2}$ replaces the bidirectional cross-attention in TDCA with a unidirectional HS→SAR path (only SAR features are updated by attending to HS features).

$\boldsymbol{C_3}$ replaces the bidirectional cross-attention in TDCA with a unidirectional SAR→HS path (only HS features are updated by attending to SAR features).

$\boldsymbol{C_4}$ replaces the complementary gated fusion with mean fusion (elementwise averaging of aligned HS and SAR features).

$\boldsymbol{C_5}$ removes the query/key enhancement from TDCA (i.e., no key refinement and no query conditioning) and keeps the subsequent cross-attention and fusion operations unchanged.

The results are summarized in Table IX. $\boldsymbol{K_1}$+ P1 + P3 and $\boldsymbol{K_1}$+ P2 + P3 are not considered since they are meaningless. The principle-level alternative models are initially analyzed to show that performance improvements are not merely due to increased model capacity. With modality-specific fidelity (P1), K_2 increases OA and AA to 90.17% and 61.50%, indicating that dedicated HS and SAR modeling effectively preserves intrinsic discriminative patterns. These accuracies are increased to 90.93% and 62.91% by adding generalization & training stability (P2), demonstrating consistent performance benefits even under limited supervision. By considering cross-modal complementary alignment (P3), the OA and PR of K_4 reach 91.06% and 72.79%, proving that semantic alignment and selective information propagation are more effective than naive fusion. Compared with K_1, K_5 improves OA and $F1$ by 3.36% and 0.1020%, respectively, showing that the superiority of GenMSLnet stems from jointly satisfying the three complementary principles, not merely from increased model capacity.

Subsequently, multiple component-level alternative models are analyzed to reflect model contributions in GenMSLnet.

1) *P1 Related Variants:* Notably, A_1 and A_2 decrease [OA, $F1$] to [92.03%, 0.6668%] and [91.98%, 0.6598%], showing that removing attention mechanisms from generators degrades both overall and balanced performance. A_4 drops AA from 67.26% to 63.13% and $F1$ from 0.6849 to 0.6434, confirming the importance of multiscale modeling for capturing textures and spatial variations. Replacing multiscale structures with stacked convolutions results in performance degradation, particularly when both branches are simplified. In G_{SAR}, A_5 replaces deformable convolutions with standard ones, decreasing OA and KC to 91.96% and 0.8834%. These results confirm that attention mechanisms, multiscale modeling, deformable sampling, and orientation enhancement strengthen modality-specific representations and improve cross-modal matching.

2) *P2 Related Variants:* In P2, regularization and stabilization mechanisms mainly affect robustness and class balance. B_1 replaces SOR with SN in generators, achieving [OA, $F1$] of [92.00%, 0.6643%], indicating that SN improves stability and fails to match SOR in robustness and class balance. When replacing SOR with ℓ_2 regularization, B_2 alleviates this decline but still underperforms K_5, suggesting that SOR provides a stronger and more suitable constraint. Discriminator simplification also leads to consistent performance degradation. B_3 decreases OA to 92.01% and further highlights the importance of the discriminator structure in ensuring stable adversarial learning. B_4 removes noise input and results in a notable decline in performance, confirming that noise remains beneficial for robustness during adversarial augmentation. These results highlight that SOR-related regularization, discriminator structure, and stability-oriented designs are contributors to robust and balanced learning.

3) *P3 Related Variants:* Cross-modal alignment is the most sensitive factor in these variants. C_1 removes bidirectional dynamic cross-attention and results in the most significant performance drop, i.e., OA drops from 92.22% to 91.40%, AA drops from 67.26% to 64.17%. When using single interaction directions, C_2 and C_3 also have suboptimal performance, indicating that both modalities should guide each other for effective complementarity. C_4 decreases AA to 64.90% and $F1$ to 0.6550 since it replaces gated fusion with mean fusion. These results prove that selective fusion is more reliable than uniform aggregation. The C_5 results suggest that the query/key enhancement helps GenMSLnet capture and align cross-modal features. These results confirm that bidirectional interaction, selective fusion, and attention enhancement are necessary elements for effective cross-modal complementarity.

3) Necessity of Both RS Modalities: Several scenarios with different RS modality pairs are considered as follows.

1) HS+∅ chooses the HSI as the input and removes the SAR-related structures.
2) SAR +∅ chooses the SAR data as the input and removes the HS-related structures.
3) HS +HS chooses the HSI as its dual inputs.

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TABLE X

CLASSIFICATION ACCURACIES, INCLUDING OA (%), AA (%), PR (%), KC (×100), AND $F1$ (×100), ON THE AUGSBURG DATASET TO PROVE THE NECESSITY OF BOTH RS MODALITIES

Scenarios	OA (%)	AA (%)	PR(%)	KC (×100)	F1 (×100)
HS+∅	83.38	56.18	60.05	76.11	54.79
SAR+∅	84.21	48.38	53.87	76.97	46.94
HS+HS	89.56	63.46	69.78	82.22	61.46
SAR+SAR	87.70	55.38	59.77	80.60	54.25
HS+SAR	**92.22**	**67.26**	**73.36**	**88.76**	**68.49**

TABLE XI

CLASSIFICATION ACCURACIES AND TRAINING TIMES PER EPOCH WITH DIFFERENT SLIDING WINDOW SIDES ON THE AUGSBURG DATASET TO SELECT THE OPTIMAL SLIDING WINDOW SIDE VALUE

Side	OA (%)	AA (%)	PR (%)	KC(×100)	F1 (×100)	Time (s)
3	91.45	65.19	71.90	87.01	65.25	**12.67**
5	**92.22**	**67.26**	**73.36**	**88.76**	**68.49**	17.98
7	91.89	66.25	72.38	88.14	67.60	25.43
9	91.02	64.48	72.04	86.74	64.99	35.19
11	90.56	63.59	70.99	86.12	62.05	44.51

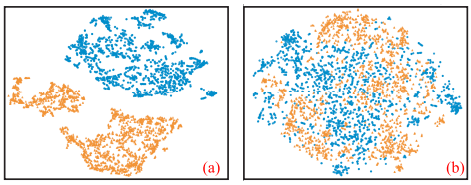


Fig. 10. T-SNE visualization of HS (blue points) and SAR (yellow points) feature embeddings (a) before MDGAN and (b) after MDGAN in GenMSLnet.

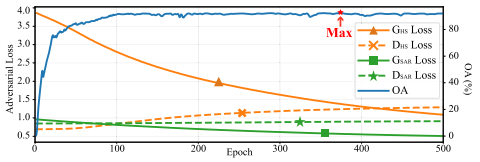


Fig. 11. Training curves for OA and several losses under the 1% training-sample setting on the Augsburg dataset to prove convergence and stability.

4) SAR +SAR chooses the SAR data as its dual inputs.
5) HS +SAR (GenMSLnet) chooses the HSI and SAR data as its dual inputs, respectively.

Table X shows that HS+SAR achieves the best performance across all metrics, confirming the necessity of jointly using HSI and SAR data. Removing either modality leads to a sharp decline; e.g., HS+SAR outperforms HS+∅ and SAR+∅ by 8.84% and 8.01% in terms of OA, respectively. The improvement is more pronounced on balanced metrics. Compared to HS+∅, HS+SAR increases AA/$F1$ from 56.18%/0.5479% to 67.26%/0.6849%. This indicates that a single modality is insufficient given the class imbalance and spectral–spatial mixing in complex capturing areas. HSI provides rich spectral features but struggles to distinguish spectrally similar classes, while SAR data provides distinct structural and textural patterns but lacks detailed spectral information. Removing either modality reduces class separability, leading to a decline in performance.

HS+HS and SAR+SAR further demonstrate that duplicating the same modality cannot capture cross-modal complementarity. Although HS+HS and SAR+SAR are relatively satisfactory, their accuracies are notably lower than those of HS+SAR, indicating only a limited improvement over their respective single-modality baselines. HS+SAR (GenMSLnet) reflects strong consistency and reliable predictions by leveraging the complementary information of HSI and SAR data.

4) Sliding Window Side Value Selection: The sliding window side value is selected from $\{3, 5, 7, 9, 11\}$, with corresponding accuracies and per-epoch training time reported in Table XI. While a larger window provides more information, excessively large windows introduce surrounding clutter and high computational consumption, which disturbs target information and hurts generalization. When the side value increases from 3 to 5, its performance improves across all metrics, while training time increases from 12.67 to 17.98 s per epoch. This result indicates that sufficient discriminant information and tolerable computational cost are balanced when the side value is 5. When the side value becomes 7, OA and AA drop to 91.89% and 66.25%, respectively. The training time rises to 25.43 s, making GenMSLnet unworthy. Performance degrades more obviously with larger windows. For side values nine and 11, OA declines to 91.02% and 90.56%, $F1$ declines to 0.6499 and 0.6205, while training time per epoch increases to 35.19 and 44.51 s. Thus, the optimal side value is set to five.

5) T-SNE Visualization of Feature Embeddings: To verify that MDGAN improves cross-modal distribution alignment, both qualitative and quantitative results are provided on the Augsburg dataset. In particular, T-SNE visualizes the HS and SAR feature embeddings, as shown in Fig. 10, while maximum mean discrepancy (MMD) is adopted to quantify the distribution discrepancy between both modalities. Smaller MMD values indicate better cross-modal consistency. Fig. 10(a) shows that the HS and SAR embeddings are clearly separated without MDGAN, revealing a notable cross-modal distribution gap. After introducing MDGAN, the modality embeddings become substantially more overlapped, as shown in Fig. 10(b), which qualitatively indicates improved alignment. Quantitatively, averaged over repeated runs, the MMD decreases from 0.2855 ± 0.0021 to 0.0101 ± 0.0006, confirming MDGAN reduces the cross-modal distribution discrepancy.

6) Convergence Behavior and Training Stability: To clarify the convergence behavior and training stability under the 1% training-sample setting, the training curves on the Augsburg dataset are provided in Fig. 11. In each epoch, a fixed two-to-one updating ratio is adopted, i.e., the generators are updated twice while the discriminators are updated once per cycle, to avoid discriminator overfitting and maintain a balanced adversarial game under scarce labels. As shown in Fig. 11, the classification accuracy, i.e., OA, increases rapidly in the early stage and then converges to a stable equilibrium (the best OA is marked). Meanwhile, both the generator losses, i.e., G_{HS} and G_{SAR}, and discriminator losses, i.e., D_{HS} and D_{SAR}, remain bounded and evolve smoothly without evident divergence or unstable oscillation, indicating stable adversarial optimization. In addition, the complete training procedure was repeated ten times with different random seeds, without encountering collapse or divergence. The relatively moderate standard deviations in Tables IV–VI further support the robustness and reproducibility of GenMSLnet under limited supervision.

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V. Conclusion

In this article, GenMSLnet has been presented for HS and SAR joint classification. The method mainly addresses the critical challenge of fusing both RS modalities by explicitly incorporating their distinct characteristics. Moreover, to mitigate their sensitivity to environmental interference, a GAN-based structure has been considered to enhance feature robustness and alignment. Our core contribution lies in the development of modality-specific feature learning strategies that adaptively optimize feature extraction according to the intrinsic properties of RS modalities. In particular, dual-stream architectures have been designed to handle HSI and SAR data separately, accounting for their distinct characteristics and properties. This architecture enables comprehensive cross-modal spectral–spatial–structural feature alignment, promoting efficient information interaction. Experiments demonstrate that GenMSLnet outperforms state-of-the-art techniques, exhibiting strong robustness and generalization ability. Ablation studies show the necessity of the presented structures and strategies. Future work will focus on improving the robustness under more complex environmental conditions.

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